Practice Paper A - Exam Solutions

1. (a) The times, in seconds, taken by 20 people to solve a simple puzzle were

$$17 \quad 19 \quad 22 \quad 26 \quad 28 \quad 31 \quad 34 \quad 36 \quad 38 \quad 39 \quad 41 \quad 42 \quad 43 \quad 47 \quad 50 \quad 51 \quad 53 \quad 55 \quad 57 \quad 58$$

(i) Calculate the mean and standard deviation of these times. (6 marks)

Solution: The mean is the sum of the numbers divided by the number of values, which is 20.

$$\bar{x} = \frac{17+19+22+26+28+31+34+36+38+39+41+42+43+47+50+51+53+55+57+58}{20} = \frac{787}{20} = 39.4$$

[2 marks]

The standard deviation is given by:

$$s_x = \sqrt{\frac{1}{n-1}\left(\sum x^2 - \frac{(\sum x)^2}{n}\right)}$$

[1 mark]

We know $\sum x = 787$ from above. We have $n = 20$, and we can find the sum of squares:

$$17^2+19^2+22^2+26^2+28^2+31^2+34^2+36^2+38^2+39^2+41^2+42^2+43^2+47^2+50^2+51^2+53^2+55^2+57^2+58^2=34023$$

[2 marks]

Then

$$s_x = \sqrt{\frac{1}{19}\left(34023 - \frac{787^2}{20}\right)} = \sqrt{160.76} = 12.68$$

[1 mark]

Later, after eating a protein bar, the same group of people were given a different puzzle to solve, and the mean solve time was 25.1 seconds, with a standard deviation of 4.5 seconds.

(ii) Comment on the meaning of these results. (3 marks)

Solution: The mean of the results after the protein bar was lower, which meant on average people solved the second puzzle more quickly. The standard deviation is also smaller, which means the results were more closely clustered around the mean and there was less variation. [3 marks]

(iii) Give two reasons why these results may not be useful in determining the effect of the protein bar on puzzle solving ability. (4 marks)

Solution: Reasons might include:

- The two puzzles may have had different difficulty levels, or been different types of puzzle
- The people may have learned something from solving the first puzzle they used in solving the second
- The people may have been more tired later on in the day

- The effect might have been the same if they had eaten any kind of food so it may not have been due to the protein bar
- There may have been other differences between two instances of solving a puzzle (room conditions etc)
- Any other reasonable suggestion

[4 marks (2 per suggestion, 1 if underexplained)]

(b) As part of a larger study, the same experiment was performed on a larger cohort, and the results were found to be normally distributed. Let $X \sim N(41, 10.5)$ represent the performance of 100 people before the protein bar. After eating the bar, the mean solve time was 38.5 seconds.

Perform a statistical test at the 5% significance level to determine whether their performance has improved significantly. State your hypotheses and perform the test. (12 marks)

Solution: We will perform a z-test since we have a large enough sample, and since we are measuring a change in the mean, we can use the distribution of sample means for X.

[1 marks]

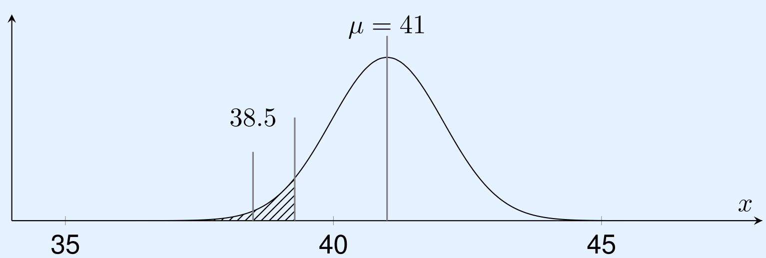
For samples of size 100, this will be given by

$$\bar{X} = N\left(41, \frac{10.5^2}{100}\right)$$

[2 marks]

We knew that the mean was 41 seconds before the protein bar, so we assume this has not changed and that people are not solving the puzzles any faster.

H_0: The mean has not changed, so $\mu = 41$
H_1: The mean has decreased, so $\mu < 41$ [2 marks]



This is a lower tail test, so for a 5% significance level, the z-score needs to be in the critical region of -1.64 or less to reject the null hypothesis. [1 mark]

For our sample, $\bar{x} = 38.5$. The z-score for this value can be calculated as:

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{38.5 - 41}{\frac{10.5}{\sqrt{100}}} = -2.38$$

[2 marks]

This is in the critical region. Alternatively, a z-score of -2.38 corresponds to a probability of $1 - 0.9913 = 0.0087$ or 0.9%, which is lower than the significance level of 5%.

[2 marks for either approach]

We can reject the null hypothesis and accept the hypothesis that the solve time has decreased.
[2 marks]

Total marks for question 1: [25]

2. (a) Find the determinant of the matrix $M = \begin{pmatrix} -1 & 2 & 0 \\ 7 & 4 & 3 \\ 5 & 2 & -4 \end{pmatrix}$ (5 marks)

Solution: Expanding along the first row:

$$\det(M) = \begin{vmatrix} -1 & 2 & 0 \\ 7 & 4 & 3 \\ 5 & 2 & -4 \end{vmatrix} = -1 \times \begin{vmatrix} 4 & 3 \\ 2 & -4 \end{vmatrix} - 2 \times \begin{vmatrix} 7 & 3 \\ 5 & -4 \end{vmatrix}$$

$$= -1(-16-6) - 2(-28-15) = 22 + (2 \times 43) = 108$$

[3 marks]

(b) (i) Calculate the dot product of the two vectors $\mathbf{u} = (2, 3)$ and $\mathbf{w} = (1, 6)$. (2 marks)

Solution:

$$\mathbf{u} \cdot \mathbf{w} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 6 \end{pmatrix} = 2 \times 1 + 3 \times 6 = 20$$

[2 marks]

(ii) Calculate the magnitude of the vector $\mathbf{u} + \mathbf{v}$. (4 marks)

Solution:

$$\mathbf{u} + \mathbf{v} = (2, 3) + (1, 6) = (3, 9)$$

[1 mark]

Then the magnitude is:

$$\|u + w\| = \sqrt{3^2 + 9^2} = \sqrt{9 + 81} = \sqrt{90} \approx 9.49$$

[3 marks]

(c) The following data was collected about the ages, and scores from the first three darts thrown, of participants in a friendly darts match.

Age (X)	Score (Y)
26	60
29	68
30	58
35	40

Calculate the population covariance matrix for this data. (12 marks)

Solution: The population variance is given by $\frac{\sum(x_i - \bar{x})^2}{n}$. [1 mark]

Here, $\mu_x = 30$, $n = 4$, so:

$$var(x) = \frac{(29-30)^2 + (26-30)^2 + (30-30)^2 + (35-30)^2}{4} = 10.5$$

We have $\mu_y = 56.5$, $n = 4$, so:

$$var(y) = \frac{(68-56.5)^2 + (60-56.5)^2 + (58-56.5)^2 + (40-56.5)^2}{4} = 104.75$$

[4 marks]

Then the covariance is given by:

$$\begin{aligned} cov(x,y) = cov(y,x) &= \frac{\sum(x_i - \mu_x)(y_i - \mu_y)}{n} \\ &= \frac{(68-56.5)(29-30) + (60-56.5)(26-30) + (58-56.5)(30-30) + (40-56.5)(35-30)}{4} \\ &= -27 \end{aligned}$$

[4 marks]

The population covariance matrix is then: $\begin{pmatrix} 104.7 & -27 \\ -27 & 10.5 \end{pmatrix}$

[3 marks for a correctly stated matrix]

Total marks for question 2: [23]

3. (a) A large office building buys pens from two suppliers - Office World, and Stapler King. The office manager knows that 10% of the pens supplied by Office World and 5% of those supplied by Stapler King are defective.

100 pens are stored together in a box, of which 65 are from Office World and the remainder from Stapler King. A member of staff selects a pen at random from the box.

(i) Find the probability that the pen is from Office World. (2 marks)

Solution: 65 of the 100 pens are from Office World, so there is a 65% or 0.65 chance of the pen being from Office World. [2 marks]

(ii) How many of the pens in the box would you expect to be both from Stapler King and defective? (3 marks)

Solution: 35 of the pens are from Stapler King. Of those, 5% are defective, so we would expect there to be $35 \times 5\% = 1.75$ defective pens from Stapler King in the box of 100. [3 marks]

(iii) Find the probability that a randomly chosen pen from the box is defective. (4 marks)

Solution: If there are 65 pens from Office World, of which 10% are defective, there should be 6.5 defective Office World pens, along with the 1.75 defective pens from Stapler King. [2 marks]

So we would expect there to be $6.5 + 1.75 = 8.25$ defective pens in total (out of 100 pens), so there is an $0.0825 = 8.25\%$ probability a randomly chosen pen is defective. [2 marks]

A member of staff who took a pen from the box found that their pen was defective, and complained to the college management about the quality of pens from Office World.

(iv) Given that the pen was defective, find the probability that it was supplied by Office World. (5 marks)

Solution: Let A be the probability of the pen being from Office World, which is 65% and B the probability that it is defective, which is 8.25%. We know that 10% of Office World pens are defective. [1 mark]

Bayes' theorem says that:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} = \frac{10\% \times 65\%}{8.25\%} = \frac{0.1 \times 0.65}{0.0825} \approx 0.787 = 78.7\%$$

[3 marks]

Given it is defective, the probability the pen is from Office World is 78.7%. [1 mark]

(b) The office manager wishes to do some more accurate modelling of the pen situation, and suggests a binomial distribution could be used to create a model for X, the number of defective pens they might expect to have out of the 10,000 pens they have ordered from Office World. Is this a suitable model, and if so how would you express it? Explain your answer. (6 marks)

Solution: In order to model a situation using a binomial distribution, we need four conditions to be satisfied:

- Two outcomes - here, these are 'defective' and 'not defective'
- Independence - each pen is manufactured separately so their chance of defectiveness is independent
- Fixed number of trials - there are 10,000 pens in the order
- Fixed probability of success - we know each pen from Office World has a 10% probability of being defective.

[4 marks]

Therefore this is a suitable situation to model using a binomial distribution, which we could write as $X \sim B(10000, 0.01)$. [2 marks]

Total marks for question 3: [20]

4. (a) It is believed that a high school student's favourite film genre is related to their favourite school subject. Data scientists wish to test this theory by surveying 500 school students from the region's 10 schools, and asking them to indicate their favourite film genre and favourite subject from a selection of options.

(i) The data scientists need to decide between collecting a simple random sample or a cluster sample. Describe what each involves, briefly indicate how you might implement it, and compare the two methods. (6 marks)

Solution: A simple random sample involves picking members of the population at random, without replacement. This could involve obtaining a list of all the students from all 10 schools, and using random number generator to pick 500 names from the whole list. [2 marks]

Cluster sampling involves splitting the population into clusters and taking a random sample from each. This could involve asking each school to pick 50 students, or getting the list of students from each school and using a random number generator to choose 50 of them. [2 marks]

The simple random sample method might not give you the same number of students from each school, so it may not be equally representative of all the schools in the region. It would be more likely to give a representative sample of high school students overall. [2 marks]

(b) The survey was conducted, and the results are indicated in the table below.

	Comedy	Action	Romance	Thriller
Maths	51	52	37	55
Sports	59	63	41	33
Geography	35	31	28	15

The data science team wish to determine whether there is any association between favourite subject and favourite film genre by using a chi-squared test for independence at the 5% significance level.

(i) State the null and alternative hypotheses for this test. (2 marks)

Solution: H_0: Favourite subject is independent of favourite film genre
H_1: Favourite subject is not independent of favourite film genre [2 marks]

(ii) Write down the number of degrees of freedom for this table. (2 marks)

Solution: $\nu = (\text{rows} - 1)(\text{columns} - 1) = (3-1)(4-1) = 2 \times 3 = 6$ [2 marks]

(iii) Calculate the chi-squared test statistic for this data. (15 marks)

Solution: The totals for each category are added to the table:

	Comedy	Action	Romance	Thriller	Total
Maths	51	52	37	55	195
Sports	59	63	41	33	196
Geography	35	31	28	15	109
Totals	145	146	106	103	500

For each combination of movie type and favourite subject, we calculate the expected values, and the value of $\frac{(O-E)^2}{E}$:

	Comedy (O)	Comedy (E)	$\frac{(O-E)^2}{E}$
Maths	51	$\frac{195}{500} \times 145 = 56.55$	$\frac{(51-56.55)^2}{56.55} = 0.545$
Sports	59	$\frac{196}{500} \times 145 = 56.84$	$\frac{(59-56.84)^2}{56.84} = 0.082$
Geography	35	$\frac{109}{500} \times 145 = 31.61$	$\frac{(35-31.61)^2}{31.61} = 0.364$

[3 marks]

	Action (O)	Action (E)	$\frac{(O-E)^2}{E}$
Maths	52	$\frac{195}{500} \times 146 = 56.94$	$\frac{(52-56.94)^2}{56.94} = 0.429$
Sports	63	$\frac{196}{500} \times 146 = 57.232$	$\frac{(63-57.232)^2}{57.232} = 0.581$
Geography	31	$\frac{109}{500} \times 146 = 31.828$	$\frac{(31-31.828)^2}{31.828} = 0.022$

[3 marks]

	Romance (O)	Romance (E)	$\frac{(O-E)^2}{E}$
Maths	37	$\frac{195}{500} \times 106 = 41.34$	$\frac{(37-41.34)^2}{41.34} = 0.456$
Sports	41	$\frac{196}{500} \times 106 = 41.552$	$\frac{(41-41.552)^2}{41.552} = 0.007$
Geography	28	$\frac{109}{500} \times 106 = 23.108$	$\frac{(28-23.108)^2}{23.108} = 1.036$

[3 marks]

	Thriller (O)	Thriller (E)	$\frac{(O-E)^2}{E}$
Maths	55	$\frac{195}{500} \times 103 = 40.17$	$\frac{(55-40.17)^2}{40.17} = 5.475$
Sports	33	$\frac{196}{500} \times 103 = 40.376$	$\frac{(33-40.376)^2}{40.376} = 1.347$
Geography	15	$\frac{109}{500} \times 103 = 22.454$	$\frac{(15-22.454)^2}{22.454} = 2.474$

[3 marks]

The χ^2 test statistic is given by $\chi^2_\nu = \sum \frac{(O_i-E_i)^2}{E_i}$:

$$0.545+0.082+0.364+0.429+0.581+0.022+0.456+0.007+1.036+5.475+1.347+2.474 = 12.818$$

[3 marks]

(iv) Write down the conclusion to the test and explain what it means. (3 marks)

Solution: Looking up in the table, for $\nu = 6$ the critical value for 5% significance is 12.592. [1 mark]

Since $12.818 > 12.592$, we can reject the null hypothesis and conclude that there is some relationship between favourite subject and favourite film genre. [2 marks]

Total marks for question 4: [33]